

Interacting with autonomous systems and intelligent algorithms – new theoretical considerations on the relation of understanding and trust

Abstract

Contemporary technology increasingly relies on artificial intelligence and machine learning and is therefore able to act increasingly autonomously. This leads to systems that display agency instead of merely serving as tools that are utilized by a human user who is in charge. Given that these autonomous systems take roles that emulate human functions (e.g., decision making, tutoring, counseling), it is increasingly important to scrutinize how humans interact with these systems and to which extent they understand and trust these systems. While recent research has specifically addressed how to foster users' understanding of algorithms, we argue that fostering calibrated trust, in the sense of trust that is warranted and calibrated to the actual reliability of the system, might be more fruitful. Building on these considerations we propose a theoretical research framework on the interrelation of human trust and understanding and discuss how they might be affected by design variables such as explanations, anthropomorphization and trust cues.

Keywords: human-machine interaction, agency, trust, understanding, anthropomorphization, explainable AI

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Technology is increasingly relying on artificial intelligence and machine learning which enables machines and systems to take on new roles in everyday life by assisting humans with their tasks and decisions. This, for example, includes people relying on their entertainment platform for recommendations, medical doctors being assisted in their diagnosis by data-driven systems, learning programs presenting tasks based on a learner's individual progress, or the pre-selection of applicants in a staffing procedure (e.g., Kunkel et al., 2021; Nensa et al., 2019; Sundar, 2020). In these cases, humans directly communicate with or interact with AI systems. Although this interaction will sometimes be more and sometimes be less explicitly verbal, the situation resembles interpersonal communication situations, in which humans receive advice from someone else. Whether users want to interact with these systems and whether they decide to, for example, take the advice will be dependent on how they evaluate the system regarding accuracy and reliability. A decisive question therefore is to what extent human users can understand the functioning of these new, intelligent systems. Building on the (normative) notion that it is better if users understand the functioning of a system, explainable AI (XAI) research has flourished in recent years (Dosilovic et al., 2018; Mohseni et al., 2020). Here, ways are sought that enable to uncover and communicate a system's functioning in a form that even laypersons can better grasp the processing. Still, research from various related areas also indicates that understanding of these complex relationships is limited (Bromme & Girth, 2021) and that not everyone wants to be informed about a machine's processing (Ngo & Krämer, 2021). Therefore, we argue that besides the prevalent approach to inform people about the technology's functioning, alternative ways of helping people to assess the reliability of a system should be explored. Based on arguments from science communication research, we suggest that epistemic trust (Sperber et al., 2010) –

in the sense of trust that is shown to experts or institutions that display competence related trust cues - could be a beneficial means to support laypeople when trying to judge algorithms. Transferred to the field of artificial intelligence, this would entail an expert institution trying to communicate the level of trust that should be warranted to a system – in the sense of calibrated trust (Lee & See, 2004). This entails that users would be informed about the degree of the system's reliability to ensure that they neither trust too much nor too little.

In line with these aspects, it further needs to be discussed and better understood how the way systems are designed influences people's understanding and trust. Here, as design variables those aspects are addressed which communicate the trustworthiness and/or the inner functioning of the system: This can be done, for example, via anthropomorphizing the system, by explaining its functions or by trust cues (e.g., trust seals). By means of empirical research, it needs to be further scrutinized how design variables like these influence trust and understanding and how these, in turn, influence important dependent variables such as intention to use and satisfaction with the system. With a view to contributing to responsible computing, it is important to note that from a normative point of view it is important that intention to use should only be increased against the background of a correct communication of the system's reliability. The present compilation of relevant variables and the fact that systems have developed towards autonomy and own agency already suggest that models merely looking at determinants for acceptance by the user (such as the Technology Acceptance Model (TAM; Davis, 1985), or the Unified Theory of Acceptance and Use of Technology (UTAUT, Venkatesh et al., 2003)) are no longer sufficient to explain human-technology-interaction.

The goal of the present paper therefore is to summarize the existing evidence on what is known about human understanding of algorithms, human trust, their relationship as well as how to foster both with a view to influence the intention to use systems and humans' satisfaction with them. Based on this, a theoretical research framework is proposed that

includes design variables, humans' trust and understanding as well as outcome variables such as the intention to use the system.

From tool to interaction partner

Contemporary technology entails an increasing degree of agency while becoming more ubiquitous and seamless (it, for instance, no longer includes some kind of apparatus with which the individual user decides to interact). Interacting with technology no longer takes the form of classic human-computer interaction; instead, humans interact with intelligent algorithms on the Internet, communicate with speech assistants such as Alexa or Siri or have decision-support systems give advice regarding complex tasks. Technology no longer merely serves as a tool but is increasingly acting in roles that we know from human-human interaction (Köbis et al., 2021). In an unprecedented way, technology in the form of intelligent algorithms and decision support systems has come to emulate functions that were previously solely dedicated to other humans: They act as counsellors, teachers or supervisors. Unlike earlier technology, which was predominantly used as a tool to support human workflow, recent algorithm- and artificial intelligence-based systems assume increasing agency – sometimes not only by supporting human decisions but also by suggesting or even executing decisions themselves. Köbis et al. (2021) aptly summarise: “As machines powered by artificial intelligence (AI) influence humans' behaviour in ways that are both like and unlike the ways humans influence each other, worry emerges about the corrupting power of AI agents.” In a comprehensive review, the authors demonstrated that algorithms are increasingly taking on roles as advisors, role models, partners or delegates, and can be influential for human behaviour and decisions. Drawing on early conceptualisations by Rammert (2008), Sundar (2020) recently described the tension between **machine agency** and **human agency** by stating that proactivity has shifted to machines that are increasingly acting as sources in their own right. While traditional technology was largely passive (i.e. depending

on human operation), semi-active (with some self-acting aspects) or reactive (systems with feedback loops, e.g., for air conditioning/climate control) (Rammert, 2008), technology has long since become more proactive (e.g., self-activating programmes such as car stabilisation). Until recently, there was often a balance between technology's and human users' agency, insofar as technology's actions were performed at the user's command. Nowadays, however, technology is becoming increasingly co-operative (with distributed and self-coordinating programmes, e.g., in smart homes). This shifts the balance of agency even more towards the system, as the human users are confronted with actions or decisions which they have neither controlled nor comprehended (Sundar, 2020). The "rise of the machine agency" is not necessarily problematic: After all, machine agency is built to support the human user in complex tasks and if there is "mutual augmentation" there is no need to demand that only the user holds agency (Sundar, 2020). With these co-operative systems, it is, however, often impossible to scrutinise what part of a decision has been taken by the technology and what part by the human, as decisions are increasingly taken by the interactions of humans and technologies in conjunction. This unified decision making of humans with computer artifacts that shift from tool to agent has been aptly described as "hybrid intelligence" and "federated social cognitive systems" (Akata et al., 2020; Rozendaal et al., 2020) – rendering it difficult for human users to adequately perceive to what degree agency lies with the machine and to what degree they are able to assert control.

In psychology, the social cognitive perspective on agency is historically and currently exemplified by the work of Bandura (1977; 2006). To be an agent means deliberately influencing one's own actions and life. Key components include intentionality, forethought, self-reactiveness and self-reflectiveness. Intentionality refers to the fact that people form intentions (plans for action, strategies), usually involving others (collective intentionality). Forethought refers to the ability to anticipate future responsibilities or outcomes in current activities. Self-reactiveness means that agents are not only planners and forward thinkers but

also self-regulators. In other words, agency includes the ability not only to make decisions and design a plan of action, but also to conceive appropriate procedures to monitor their execution. Finally, self-reflectiveness refers to the ability of agents to monitor both actions and themselves. Agency has long been conceptualised as a uniquely human characteristic, as can best be exemplified by a quote from Bandura (2006): "The metacognitive capability to reflect upon oneself and the adequacy of one's thoughts and actions is the most distinctly human core property of agency." Nowadays, the question of whether technology can (objectively) be an agent in the sense as defined by Bandura (2006) has been answered in the affirmative. It has been stated that technology no longer purely serves as a tool but can increasingly be likened to an agent (Akata et al., 2020; Rozendaal et al., 2020). Along similar lines, other concepts refer to technology increasingly exhibiting control (Shneiderman, 2020) or engaging in proactivity (Isbell & Pierce, 2005). From a social science point of view these developments raise the question how humans interact with these new kinds of agents. With a view to the usage of technology in everyday life, the ultimate question is what the prerequisites are for humans to decide to use the systems (if this is free choice) and whether to accept the decisions that are suggested by the technology. Two prerequisites for this kind of acceptance that are frequently discussed in the literature are understanding of the systems and trust.

What is machine “intelligence”?

In order to discuss to what degree humans can understand these new forms of automated systems which increasingly emulate human roles, decisions and (cognitive) functions, it first needs to be clarified to what degree automated systems which are developed based on machine learning are similar to or different from human cognitive functioning. Then, an overview on what is known about how humans build an understanding of these systems is given.

It is indeed difficult to assess and describe the exact functioning of automated systems that are developed via machine learning. The current chapter, therefore, does not try to give a comprehensive account on what is known about the functioning of different systems (potentially built by various forms of machine learning) but wants to give a rough orientation of why the emulation of cognitive functions by machines might differ considerably from human cognitive functioning – when perceiving objects, taking decisions or choosing behavior.

Contemporary machine learning largely relies on big data procedures to train an algorithm or system. A prominent example is training an algorithm to distinguish pictures of glassware and bottles. Here, a large sample of pictures that have been manually annotated by human coders is used as training material until the system has “learned” which pictures show a drinking glass or a bottle and can reliably categorize pictures that are new. However, the algorithm itself does not display which cues it uses to distinguish. It can only be assumed that it relies on similar cues that humans use and learned in their socialization – which would be that bottles are characterized by coloured glass, diminution to a neck, cone-shape whereas drinking glasses are characterized by transparent glass, a larger opening etc. Most systems, however, never reach quite the accuracy that humans have in distinguishing these simple forms (see, for example, the famous dog or muffin challenge¹). Also, the fact that these algorithms can often be tricked indicates that there might be different processing going on compared to what happens in human brains. Here, well-documented failures are the so-called adversarial examples. When trying to recognize a picture, changing only one pixel (that is not noticeable to human observers) might lead the algorithm to recognize an elephant instead of a living room. Research by Eykholt et al. (2018) impressively shows that small art stickers on a stop sign can cause a neural network to misclassify the sign as a speed limit 45 sign or yield

¹ <https://www.freecodecamp.org/news/chihuahua-or-muffin-my-search-for-the-best-computer-vision-api-cbda4d6b425d/>

sign. This indicates that the processing is completely different from the semantic-based recognition which is prevalent in humans. Based on this, the question arises to what degree a human user is able to grasp the differences in processing and to understand the functioning of the algorithm.

The research area of explainable AI (XAI) addresses people's ability to understand the functioning of algorithms and has recently garnered huge interest (for an overview see Dosilovic et al., 2018; Mohseni et al., 2020). Many computer science researchers have developed procedures to re-engineer what actually happens within the machine learning process and to transfer this to explanations. In recent years, the ACM FAccT network has formulated the Fairness Accountability Transparency principles (Friedler & Wilson, 2018). Of these three principles, transparency is most closely related to understanding, as it refers to the possibility to understand the system's actions in a specific case. Transparency should not merely be regarded from an expert's point of view; rather, comprehensibility is also important for end users when they are affected by the system's decision (e.g., when the system grants or rejects a loan application), or when the system's purpose is to provide decision support. But to what degree are laypersons able to comprehend the functioning of an artificial intelligence system?

Human understanding of machine intelligence

Given that automated systems function quite differently from human cognition, it is important to ask to what degree lay users of technology can comprehend the functioning of machines and algorithms. As first studies indicate that humans often lack a profound understanding, it also needs to be scrutinized why understanding fails.

Evidence on what people know and understand

Empirically, it is not trivial to measure people's understanding: According to Lai et al. (2021), objective measures span from asking participants to simulate the model's prediction, to quizzes, to detection of errors, or to asking participants to deliver a correct description themselves. Since these measures are complex to develop and apply, understanding has most often been assessed using subjective self-report measures (Anik & Bunt, 2021; Cai et al., 2019; Wang & Yin, 2021). However, studies have already demonstrated that subjective and objective measurements of understanding do not necessarily align. Most often, this is due to illusory confidence, which leads to the belief that one has understood more than is actually the case (Chromik et al., 2021).

In line with trying to scrutinize how people make sense of what is happening in automated systems, studies try to assess people's mental models (Kunkel et al., 2021) or folk theories (DeVito et al., 2018). Recent empirical findings on the extent of humans' knowledge about algorithmic functioning suggest that people only have rather superficial knowledge (DeVito et al., 2018; Kunkel et al., 2021; Ngo et al., 2020). For example, when asked to draw a figure on the data processing within a recommender system, one participant was merely able to draw a multi-armed squid, indicating their understanding of a system that relies on the collection of a multitude of data (Ngo et al., 2020). Kunkel et al. (2021) identify distinct user types who vary regarding the detailedness and (supposed) accurateness of their mental models. Other research demonstrates that people's conception is heavily influenced by what is displayed in the media (DeVito et al., 2018; Horstmann & Krämer, 2019).

Why does understanding fail?

In interpersonal human-human communication it is well known that human abilities such as perspective taking (Fussell & Krauss, 1992), Theory of Mind (Premack & Premack, 1995) and mentalizing (Frith & Frith, 2006) enable people to reconstruct the mental

processes, beliefs, emotions and attitudes of other people. While this is incredibly helpful to develop joint goals and a joint understanding with fellow humans (Tomasello, 2010), this must necessarily fail when trying to understand the “minds” of machines as the semantics of machine “intelligence” is different from human intelligence even with approaches like neural networking which tries to emulate biological intelligence (Iliadis et al., 2020). In this sense, in popular science, artificial intelligence has been likened to aliens coming to the earth which would be met by similar inability of immediate understanding (Lesch & Schwartz, 2020). But while when encountering an alien, people might not try to apply Theory of Mind techniques as the “otherness” might be rather salient (e.g., differences in colour or shape), human users will be tempted to apply procedures and techniques of human-human interaction to automated systems – as long as they execute tasks that are similar to human functioning (deciding, advising, recommending, guiding).

What might further aid to these dysfunctional tendencies, is the well-documented human tendency to react socially to technology which entails social cues (e.g., the filling of roles known from interpersonal communication, such as advisor, tutor, coach). This assumption can, first, be derived from media equation assumptions (Krämer, 2008; Nass & Moon, 2000; Reeves & Nass, 1996), which has demonstrated that people treat computers as social beings – even though they do not think that the computer warrants social treatment.

This can, second, be further aggravated by anthropomorphizing the machine further. It has long been described, for instance by Epley et al. (2007), that people are prone to humanise non-human agents, including AI-enabled technology (Li & Suh, 2021; Blut et al., 2021). According to Epley et al. (2007), based on the desire to explain and predict the behaviour of the non-human agent, knowledge about oneself or humans in general is applied - specifically when no other knowledge is available. In this line, research has demonstrated that humanisation increases when the technology seems to be unpredictable (Eyssel et al., 2011; Waytz, Morewedge et al., 2010). This kind of anthropomorphization (specifically when it is

additionally fostered by making the machine appear human-like, see below) is problematic because it might lead to the increase of perceived understanding but not necessarily to more actual understanding.

Fostering understanding with XAI

One recent development in order to assist laypersons with understanding algorithms and machines is explainable AI (XAI). The research is manifold and focusses either on making machine learning procedures more transparent for experts who initiated the machine learning or addresses how laypeople can be informed about the functioning of the particular algorithm or system. Regarding the latter, studies in this field have begun to compare the effects of different forms of explanations and to investigate the efficiency of visual examples (Cai et al., 2019; Papenmeier et al., 2022). However, it has already been pointed out that explainable AI needs to become more user- and human-centred in order to actually be helpful (Liao et al., 2020) and to truly inform users in accordance with their legal rights (Edwards & Veale, 2017). First approaches in this direction have attempted to consider human cognitive conditions and patterns (Wang et al., 2019) and to quantify what makes explanations human-interpretable (Lage et al., 2018). Still, it is difficult to teach the functioning to people who have a limited understanding of algorithmic procedures. In addition, users sometimes do not want to be informed at all (Springer & Whittaker, 2020; Ngo & Krämer, 2021) – partly because they fear that increased knowledge about, for example, the storage and processing of personal data would keep them from using the system in future.

Altogether, the state of the art suggests that the understanding of algorithmic functioning in systems is limited – both because it is difficult to explain the machine's procedures to a sufficient degree and due to the fact that even when the procedures are explained, humans will tend to apply their immediate understanding of other agents' actions (Theory of Mind) to machine functioning. A current analysis in the field of science

communication similarly suggests that understanding complex phenomena is difficult: Bromme and Gierth (2021) argue that the public understanding of science and scientific developments is bounded in the sense that for large parts of the general public, their individual understanding is limited. Constraints of understanding result from the potentially unlimited depth of scientific explanations and the complexity of the methods and procedures involved. This is even more important when it comes to conflicting scientific knowledge claims or when the scientific knowledge is still uncertain. In this case, the public needs to decide which of the conflicting knowledge claims are true for the wider public. Bromme and Gierth (2021) argue that: “most members of the general public are not able to decide by virtue of their own understanding of the relevant topic which scientific knowledge claim should be adopted as a true belief (in the sense of being justified scientifically).”

Therefore, although the majority of current research is targeted at understanding and transparency, we suggest that one should additionally explore alternative routes that help users to assess the reliability of systems.

Alternative ways to understanding: Can trust help to assess reliability?

In sum, to purely rely on understanding in the area of human-machine interaction is problematic for two reasons: first, due to laypersons’ partial **inability** to understand and, second, due to partial **unwillingness** to acquire in depth knowledge about algorithms. The first is grounded in both, the fact that “public understanding” is limited because of the huge amount of knowledge to acquire (Bromme & Girth, 2021) which leads to the difficulty to communicate the mechanisms in XAI approaches and in the fact that humans have difficulties to automatically deduce the functioning of machine intelligence because of the lack of a “theory of artificial minds”. Second, empirical research demonstrates that people sometimes are even unwilling to acquire a deeper understanding, for fear of not wanting to use the

service any longer when learning about data storage and data processing (Ngo & Krämer, 2021).

One alternative route to ensure that people can assess the reliability of the system might be trust. This drift from understanding to trust is already suggested in the field of science communication: Here, it has been suggested that the relevant question could be changed from “What to believe?” to “Whom to believe?” (Bromme & Kienhues, 2014; Bromme & Girth, 2021). Although this form of “epistemic trust” (Sperber et al., 2010) seems to be at odds with scientific reasoning and argumentation, it is the only chance for those who cannot evaluate or understand the scientific evidence or the engineering procedures and mechanisms for themselves. Still, this does not need to be seen as a denial of rationality: The topic of this reasoning is no longer seen as science as a cognitive structure but rather science as a social system. The question of “Whom to believe?” in the sense of whom to trust is still difficult and complex to answer (see also the reasoning in the unimodel of persuasion (Kruglanski & Thompson, 1999) which states that deciding whether a source is credible is functionally equivalent with the elaboration of whether to believe complex arguments). In the area of trust in artificial intelligence this would be equivalent to (effortfully) processing whether trust cues (e.g., regarding the source in the sense of the developer or of a testing institute) should be followed. Despite the obvious complexity of these trust decisions, Bromme and Gierth (2021) argue that citizens’ capability to make judgments of trust is less bounded than their capability to judge scientific claims and truly understand scientific content.

Independent of these arguments from science communication research, trust research has flourished in the area of artificial intelligence in recent years: In psychology and communication science, research has focused in particular on the degree of trust people display towards technology and intelligent algorithms – leading to conflicting findings. On the one hand, it was demonstrated that people may trust intelligent algorithms and machines less than humans, even when the algorithm outperforms the human (algorithm aversion, Dietvorst

et al., 2015). For instance, Prahl and Van Swol (2017) reported that after receiving bad advice, trust in automated advice decreased more than trust in advice from humans. On the other hand, studies have demonstrated that people rely more on algorithmic predictions than on predictions by humans – except for when people have their own experience in forecasting (algorithm appreciation, Logg et al., 2019). Further research has scrutinised how to conceptualise trust towards AI systems, under what conditions people trust machines and intelligent technology, and which factors are influential in this respect (for an overview see Bonnefon & Rahwan, 2020; Glikson & Woolley, 2020; Jacovi et al., 2021).

Other research has demonstrated that machines are trusted less when decisions have a moral component (Bigman & Gray, 2018). This, however, depends on the values embedded in the machines (Freedman et al., 2020), which are individually and culturally diverse (Awad et al., 2020). Castelo et al. (2019) showed that algorithm aversion is domain-dependent: Algorithms are trusted less for subjective (vs. objective) tasks. Based on a review of the research, Burton et al. (2020) concluded that the causes of algorithm aversion are manifold and that resolving it requires an updated research programme that focuses on theory integration. Altogether, a major shortcoming of this research strand is that it often relies on experimental vignette studies or surveys, and is only rarely based on actual interaction with technology. The relation of agency, and the perception thereof, with trust has barely been examined to date. In a theory paper, Puranam and Vanneste (2022) argue that perceived agency increases competence and trust. Nevertheless, perceived agency can backfire if betrayal is feared. Also, if the system itself does not appear to be agentic, trust is instead influenced by attributions of trustworthiness to the designer of the system.

Towards the concept of calibrated trust

Trust has long been described as an important variable in human-technology interaction (Lee & See, 2004). Recently, the VDE has published a new application guide for

trustworthy AI ([VDE-AR-E 2842-61-2](#)), which outlines the means by which an AI system must guarantee safety, information security, usability or ethical and regulatory issues. Accordingly, the guide also defines requirements that contribute to trustworthiness. Besides these truly technical papers, it is broadly acknowledged that due to the growing complexity of automation and the impracticality of achieving a complete understanding, people rather react socially to machines – meaning that trust, rather than understanding, influences whether people rely on the technology (Lee & See, 2004). Nowadays, research in this field is increasingly interested in how human users might develop trust when interacting with AI technologies. Here, two central observations have been made: (1) users sometimes resist using AI due to a lack of trust, while (2) users sometimes also display an overreliance on AI. Originating in earlier works on automation (Lee & See, 2004; Madhavan & Wiegmann, 2007; Parasuraman & Riley, 1997), trust has become the central variable to explain both resistance to use AI (disuse) and overreliance on AI (misuse).

Hence, many studies have aimed to adapt the notion of trust bestowed in a system to reflect the trustworthiness of systems, arriving at the concept of *calibrated trust* (Lee & See, 2004). Crucial to calibrated trust is the differentiation between the perceived and the actual trustworthiness of a system, as AI systems can be more or less *trustworthy* due to their functionality and reliability, while users might *perceive* these systems differently. In the case of calibrated trust, the perceived trustworthiness of a system matches the system's actual trustworthiness, and the user neither trusts the system less than it deserves (distrust) nor more than is actually warranted (overtrust).

In order to calibrate trust, either initial trust calibration or dynamic trust calibrations can be employed. Initial trust calibrations aim to align the user's perception of the system's trustworthiness with its actual trustworthiness as early as possible in the interaction. One way to achieve this is to inform users about a system's capabilities and limitations prior to interacting with the system. To that end, Khastgir et al. (2018) found that increasing

knowledge about a system's capabilities and limitations increased trust in both high- and low-capability autonomous vehicles. Similar results were obtained by Kraus et al. (2019), who provided participants with information about the system's reliability and reputation. Moreover, Kraus and colleagues (2019) also found that providing prior information affected users differently depending on their need for cognition.

In contrast to initial trust calibration strategies, dynamic trust calibrations allow users to gain these insights into the system's capabilities and limitations *while* engaging with a system. This is achieved, for example, by presenting information about a system's confidence throughout the interaction (McGuirl & Sarter, 2006; Okamura & Yamada, 2020; Zhang et al., 2020) or increasing a system's transparency through explanations (Poursabzi-Sangdeh et al., 2021; Zerilli et al., 2022). However, findings by Poursabzi-Sangdeh et al. (2021) also suggest that providing users with more information (through increased interpretability and transparency) decreases users' capabilities to detect and correct mistakes by the system. The authors proposed that this might be the result of information overload. Moreover, previous research has found that only providing explanations can increase trustworthiness perceptions but can also increase overreliance (Bussone et al., 2015). Papenmeier et al. (2022), however, demonstrated that the relation between the accuracy of the system, explanations given, and resulting trust is complex and that it is important to differentiate between self-reported and behavioural trust.

One important prerequisite to build a system that adequately calibrates trust (e.g., by communicating its agency honestly by means of the system's affordances) is to reliably measure people's trust and relate this to the system's functionality. There is a wealth of different – and often considerably divergent – measures to assess trust (see Lai et al., 2021, for a comprehensive overview), although only a minority of the measures follow theoretical considerations, such as the questionnaire by Körber (2019). Especially popular are measures

that rely on the ABI framework (Mayer et al., 1995), which distinguishes between the three trust dimensions ability, benevolence and integrity. Numerous self-report measures have been developed based on this framework. Trust has also been measured objectively by observing how much people's actions are influenced by the technology and how much they accept a system's suggestions (for a list of studies employing these measures, see Lai et al., 2021). Lai et al. (2021) criticised that there is a lack of common measurements and that not only do the construct and content vary across different measures, but so too is the form heterogeneous (single items, multi-item, differentiated by dimensions or not). Therefore, in sum, it is still open whether trust (especially calibrated trust) can help people to assess the reliability of a technological system.

Fostering (calibrated) trust

If - based on the considerations above – it is desirable to optimize calibrated trust, the question is how trust can be adequately calibrated and how this can be communicated. For this, institutions will be needed which check a systems reliability and communicates it. The resulting trust is epistemic trust as has been described above. Here, similarly to what has been described regarding trust in scientists, three dimensions of trust can be distinguished: expertise, integrity, and benevolence (Hendriks, Kienhues, & Bromme, 2015). The institution communicating the amount of warranted trust for a specific system therefore has to be perceived as high in competence, as honest as well as well-meaning.

A potential solution for the communication of calibrated trust to the public could be certification procedures which lead to an “AI-seal” (Paaß & Hecker, 2020). Here, current research is mainly technical and rather focusses on the criteria to check the technology (Cremers et al., 2019; Paaß & Hecker, 2020). There is, however, no research on the psychological effects of these seals and whether they can subjectively and objectively assist

people in reliably assessing the trust that is warranted for a specific system. The process can be likened to procedures of admission of new vaccines. Here, numerous people probably decided to receive a vaccination with mRNA vaccines against Covid-19 because they trust in the admission procedure and institution (e.g., the EMA) – which try to ensure calibrated trust by meticulously testing and communicating the benefits and risks of the vaccine. Whether something similar can be established in the area of artificial intelligence should be subject of future research.

Apart from these procedures and potential AI seals, it can be discussed whether intentionally fostering anthropomorphisation will be helpful. While it has been shown that anthropomorphisation (Waytz et al., 2014) as well as social behaviour (Rahwan et al., 2020) increases trust, it might be a dangerous to take the path of further instilling human-likeness as it will lead people to assume an apparent similarity between themselves and the technology and impede the understanding of the “non-human”, “alien” intelligence. Also, employing social cues can be problematic as they might lead to unwarranted trust. Independent of how trust is communicated, the most important aspect is that trust must be calibrated (Lee & See, 2004) and that the goal is to not increase trust independent of the extent to which it is warranted.

Conclusion

We hope to have clarified that the more prominent way of fostering understanding is not the only possibility of assisting humans to assess the reliability of autonomous systems. In addition, trust – in the form of epistemic, calibrated trust – needs to be considered. Still, it has become apparent that trust and understanding are probably closely connected (e.g. because studies demonstrated that explanations and, by virtue these, more transparency and understanding result in greater trust, see Poursabzi-Sangdeh et al., 2021; Zerilli et al., 2022).

To draw conclusions from the approaches and evidence discussed above, we suggest a theoretical framework which summarizes the most important variables and hypothesizes potential relations. In the center of the framework are trust and understanding including a potential mutual influence. The framework also includes design variables (trust cues (e.g., seals), anthropomorphisation and explanations) by which trust and understanding – and eventually intention to use – can be influenced.

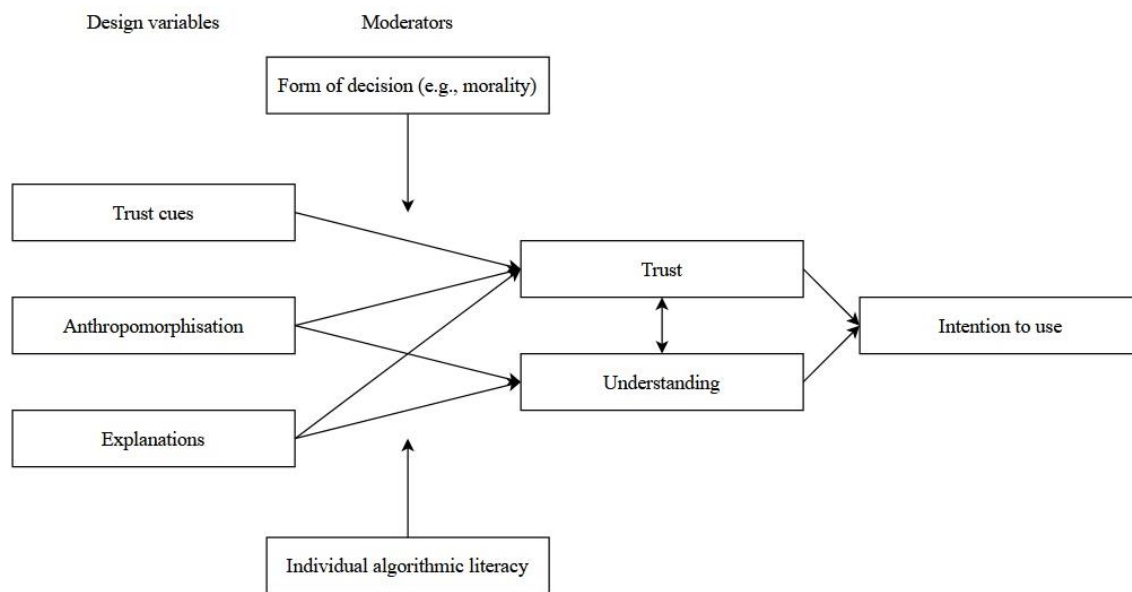


Figure 1: Theoretical research framework for future research on trust and understanding

As an example, we also include moderating variables such as the form of the decision taken (see the finding that machines are trusted less when decisions have a moral component, Bigman & Gray, 2018) as well as individual algorithmic literacy.

To use this framework to plan and conduct empirical research will help to understand the relation between the variables in order to inform fundamental research in human-machine interaction. Empirical research based on the framework will also advance applied research by helping to decide which design aspects will best foster understanding and calibrated trust. For example, it can and should be scrutinized whether fostering anthropomorphisation (Waytz et

al., 2014), leads to overtrust and/or whether it might also help to increase understanding. Similarly, it needs to be better understood how explanations might be able to influence both, understanding and trust.

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